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The Fall of the Labor Share and the Rise of Superstar Firms

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1 Big picture

- **Question.** Why has the labor share fallen in the United States and many other advanced economies, and what role did rising concentration and superstar firms play?
- **Core hypothesis.** If globalization, technology, or tougher product-market competition shifts sales toward the most productive firms in each industry, then product-market concentration rises. Since the largest and most productive firms have higher markups and lower labor shares, aggregate labor share falls.
- **Main mechanism.** The decline in aggregate labor share is not mainly because every firm reduces its own labor share. It is largely because economic activity reallocates toward firms with already low labor shares.
- **Seven predictions.** The superstar-firm hypothesis predicts rising concentration, a negative concentration-labor-share relationship, a dominant between-firm reallocation component, stronger reallocation in concentrating industries, faster productivity growth in concentrating industries, aggregate markups rising more than typical-firm markups, and similar patterns internationally.
- **Data.** The paper uses U.S. Economic Census microdata from 1982 to 2012 and international firm/industry data from OECD and CompNet sources.
- **Sectors.** The core U.S. evidence covers manufacturing, retail, wholesale, services, utilities and transportation, and finance.
- **Core empirical fact.** Sectors and industries with the largest rise in sales concentration also experience the largest fall in labor share.
- **Decomposition result.** In manufacturing, the fall in labor share is driven largely by reallocation among surviving firms toward low-labor-share firms, plus exit of relatively high-labor-share firms.
- **Markup result.** Aggregate markups rise more than the typical firm's markup because market shares reallocate toward high-markup firms.
- **Takeaway.** The paper reframes the fall of labor share as an industry-structure and firm-reallocation phenomenon rather than solely a factor-substitution, globalization, or labor-market-institution story.

2 Incremental contribution of the paper

2.1 Contribution relative to macro labor-share explanations

- Earlier explanations emphasize declining relative capital prices, capital-labor substitution, globalization, outsourcing, union decline, minimum wages, privatization, or intangible capital measurement.
- This paper adds a firm-level reallocation mechanism: even if firm-level technologies and factor shares are stable, aggregate labor share can fall when market share shifts toward low-labor-share superstar firms.
- The paper therefore makes labor share a question about the distribution of firm size, markups, and market shares within industries.

Interpretation: The key incremental step is moving from macro aggregates to firm microdata. The paper shows that aggregate labor-share decline is partly an aggregation problem: who produces output matters as much as how each producer chooses inputs.

2.2 Contribution relative to industrial organization

- IO studies document concentration and market power but often focus on prices, markups, entry, or welfare.
- This paper links concentration to functional income distribution: rising concentration predicts a falling labor share.

- It also distinguishes market power explanations from pure scale/productivity explanations by showing that concentrating industries exhibit stronger productivity growth and rising aggregate markups.

Interpretation: The paper integrates IO and macro distribution. Concentration matters not only for consumers and firms, but also for the split of income between labor and capital.

2.3 Contribution relative to firm heterogeneity and trade

- Firm heterogeneity models often show that trade or technology reallocates activity toward more productive firms.
- Autor et al. connect this logic to labor-share dynamics: more productive firms have lower labor shares because their markups are higher or because they are less labor intensive conditional on value added.
- They also show that import exposure alone does not explain the broad labor-share decline, because similar patterns appear outside manufacturing and outside traded sectors.

Interpretation: The paper uses the superstar-firm logic to connect models of selection and reallocation to macro income shares.

2.4 Contribution relative to decomposition work

- Prior decomposition work separates within-firm and between-firm changes in productivity or labor shares.
- This paper uses dynamic Melitz-Polanec decompositions to show that the falling labor share is mainly explained by between-survivor reallocation and exit.
- It then links the reallocation component directly to changes in industry concentration.

Interpretation: The decomposition is not merely accounting. It supports the mechanism that superstar reallocation, rather than uniform firm-level decline, is central to the fall in labor's share.

3 Data and measurement

3.1 U.S. Economic Census microdata

- The U.S. analysis uses Economic Census microdata from 1982 to 2012.
- The main sectors are manufacturing, retail, wholesale, services, utilities/transportation, and finance.
- The data identify establishments and firms, allowing the authors to aggregate establishments to firm-level industries.
- The Economic Census avoids several problems in macro data, including self-employment imputation and household housing treatment.

Interpretation: Census microdata make it possible to distinguish falling labor share inside firms from reallocation across firms.

3.2 Labor share measures

The paper uses two main labor-share measures:

$$LS_{jt}^{VA} = \frac{Payroll_{jt}}{ValueAdded_{jt}}, \quad (1)$$

$$LS_{jt}^{Sales} = \frac{Payroll_{jt}}{Sales_{jt}}. \quad (2)$$

- Payroll over value added is the preferred measure where value added is available, especially manufacturing.
- Payroll over sales is available more broadly across sectors.
- A broader compensation measure including fringe benefits is used in robustness tests.

Interpretation: The fall in labor share is visible in multiple definitions, reducing concern that the result is driven by one accounting convention.

3.3 Concentration measures

The paper constructs industry-level concentration using firms' sales shares:

$$CR4_{jt} = \sum_{i \in Top4(j,t)} s_{ijt}, \quad (3)$$

$$CR20_{jt} = \sum_{i \in Top20(j,t)} s_{ijt}, \quad (4)$$

$$HHI_{jt} = \sum_i s_{ijt}^2. \quad (5)$$

- Concentration is calculated at consistently defined four-digit industries.
- The authors also calculate employment-based concentration measures.
- Concentration changes are studied over five-year and ten-year differences.

Interpretation: The superstar-firm hypothesis predicts that sales concentration should matter more than employment concentration, because the mechanism is reallocation of sales and value added toward dominant producers.

3.4 International data

- EU KLEMS is used to document international labor-share trends.
- CompNet and OECD-style firm-level statistics are used to test concentration-labor-share relationships internationally.
- The international analysis is less detailed than the U.S. Census analysis but provides external validity.

Interpretation: The international evidence shows that the U.S. facts are not unique, although data limitations make the U.S. microdata the core empirical source.

4 Model and hypotheses

4.1 Firm labor share and markups

In a simple model with variable labor and a firm-specific markup μ_i , firm i 's labor share can be written schematically as

$$S_i = \frac{wL_i}{P_iY_i} = \frac{\theta_i^L}{\mu_i}, \quad (6)$$

where θ_i^L is the output elasticity of labor. Holding technology fixed, a higher markup lowers the firm's labor share.

Interpretation: Superstar firms can have low labor shares because they have high markups, high productivity, or business models that scale revenue with relatively little labor.

4.2 Aggregate labor share

Aggregate labor share in an industry is a sales- or value-added-weighted average of firm labor shares:

$$S_{jt} = \sum_{i \in j} \omega_{ijt} S_{ijt}, \quad (7)$$

where ω_{ijt} is the firm's share of industry value added or sales.

If market share moves toward firms with low S_{ijt} , then S_{jt} falls even if average firm labor share does not fall.

Interpretation: This equation is the heart of the superstar mechanism. Aggregate shares can change through reallocation, not just through within-firm changes.

4.3 Seven empirical predictions

The paper tests seven implications:

1. industry concentration rises over time;
2. industries with larger increases in concentration have larger labor-share declines;
3. the fall in labor share is driven mainly by reallocation, not by uniform within-firm labor-share declines;
4. the reallocation component is strongest in industries where concentration rises most;
5. concentrating industries exhibit faster productivity growth;
6. aggregate markups rise more than the typical firm's markup;
7. similar patterns appear outside the United States.

Interpretation: The paper's contribution is cumulative. Any one prediction could have multiple explanations, but the full pattern is tailored to the superstar-firm interpretation.

5 Empirical specifications

5.1 Industry-level concentration and labor-share changes

The baseline regression is

$$\Delta S_{jt} = \beta \Delta \text{CONC}_{jt} + \tau_t + u_{jt}, \quad (8)$$

where ΔS_{jt} is the change in labor share and ΔCONC_{jt} is the change in concentration.

- Regressions are estimated over five-year and ten-year differences.
- Industries are weighted by initial industry size.
- Standard errors are clustered by industry.
- The authors test CR4, CR20, and HHI concentration measures.

Interpretation: A negative β means that concentration growth is associated with labor-share decline.

5.2 Controls and robustness

The paper adds controls for:

- value added or sales per worker;
- industry trends;
- 1982 industry employment;
- import exposure from China;
- capital per worker;
- initial concentration.

Interpretation: The concentration-labor-share relationship remains negative after controlling for standard alternative explanations.

5.3 Melitz-Polanec decomposition

For a set of firms, aggregate labor share is decomposed as

$$S = \bar{S} + \sum_i (\omega_i - \bar{\omega})(S_i - \bar{S}). \quad (9)$$

The dynamic decomposition separates the change in labor share into:

$$\Delta S = \Delta \bar{S}_{survivors} + \Delta Reallocation_{survivors} + Exit + Entry. \quad (10)$$

- The unweighted survivor component captures within-firm average changes.
- The survivor reallocation component captures market-share shifts among continuing firms.
- Exit and entry capture composition changes from firms leaving and entering.

Interpretation: The paper finds that reallocation among surviving firms, and exit of higher-labor-share firms, explain much of the manufacturing labor-share decline.

5.4 Markup estimation

The paper estimates markups using an accounting approach and production-function approaches. The accounting approach uses the ratio of sales to variable costs, while production-function methods use first-order conditions relating input elasticities to revenue shares.

A generic markup expression is

$$\mu_{it} = \frac{\theta_{it}^V}{\alpha_{it}^V}, \quad (11)$$

where θ_{it}^V is the output elasticity of a variable input and α_{it}^V is that input's revenue share.

Interpretation: If market share shifts toward high-markup firms, aggregate markups rise even when the median firm markup changes little.

6 Identification and interpretation

6.1 What identifies the main empirical relationship

- The paper compares industries over time.
- Identification comes from differences across industries in the evolution of concentration and labor share.
- The main claim is descriptive and mechanism-oriented rather than a narrow causal estimate of concentration on labor share.
- The combination of regressions, decompositions, markups, and international patterns supports the superstar interpretation.

Interpretation: The paper is careful to present evidence as empirical support for a mechanism rather than as a randomized experiment.

6.2 Alternative explanations

- Falling capital prices could lower labor share if capital and labor are strong substitutes.
- Import competition could reduce manufacturing employment and labor share.
- Declining unions or labor-market institutions could reduce labor's bargaining power.
- Intangible capital and accounting measurement could alter value-added shares.

Interpretation: The paper does not deny these forces. It argues that none of them alone explains the full firm-level reallocation and concentration pattern.

6.3 Limitations

- Concentration is measured within four-digit industries, which may not exactly match antitrust product markets.
- The U.S. Census microdata are confidential and not directly replicable outside secure data environments.
- International evidence is less detailed and less homogeneous across countries.
- The paper provides a mechanism, but it does not fully decompose the role of technology, globalization, scale economies, and changes in competition policy as underlying causes of rising concentration.

Interpretation: The results establish a strong empirical pattern: labor-share decline is closely linked to reallocation toward concentrated, high-productivity, low-labor-share firms.

7 Tables and figures

The visuals below are directly cropped from the paper. Adjacent figures and tables are paired on the same row wherever possible.

7.1 Figure I and Figure II: broad labor-share facts

Read:

- Figure I shows that labor shares fall in many OECD countries.
- Figure II shows the decline in U.S. manufacturing labor share using payroll over value added, total compensation over value added, and payroll over sales.
- The manufacturing labor share falls sharply after 2000.

Economic message: The labor-share decline is broad, not a narrow artifact of one country or one measure.

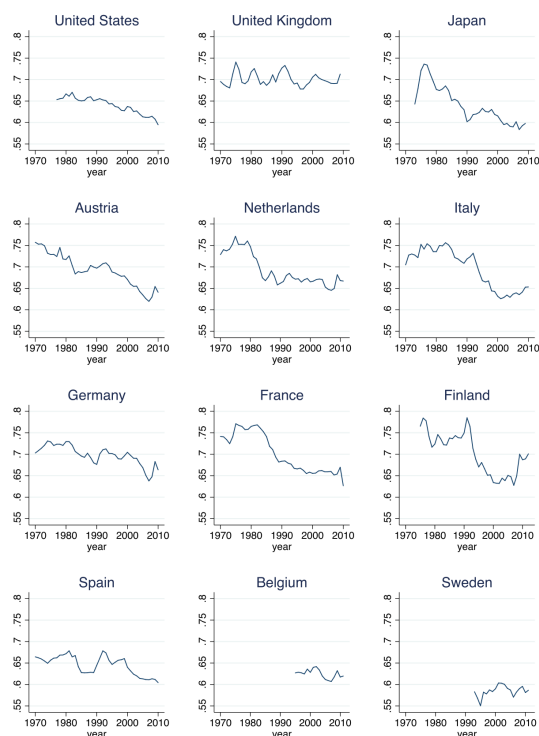


FIGURE I

International Comparison: Labor Share by Country

Each panel plots the ratio of labor compensation to gross value added for all industries. Data are from EU KLEMS July 2012 release.

Figure I: international labor share

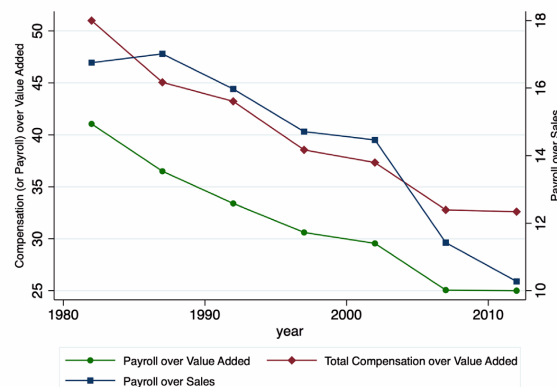


FIGURE II

The Labor Share in Manufacturing

This figure plots the aggregate labor share in manufacturing from 1982 to 2012. The green circles represent the ratio of wages and salaries (payroll) to value-added (plotted on the left axis). The red diamonds include a broader definition of labor income and plots the ratio of wages, salaries, and fringe benefits (compensation) to value added (also plotted on the left axis). The blue squares show wages and

Figure II: U.S. manufacturing labor share

7.2 Figure III and Figure IV: sector labor shares and concentration

Read:

- Figure III shows payroll-to-sales ratios by major sector.
- Manufacturing, retail, wholesale, services, and utilities/transportation experience declines, while finance is more unusual.
- Figure IV shows rising concentration across major sectors using CR4, CR20, and sales and employment weights.

- Concentration rises in manufacturing, retail, wholesale, services, utilities/transportation, and finance.

Economic message: The key facts move together: labor shares fall and concentration rises across most major sectors.

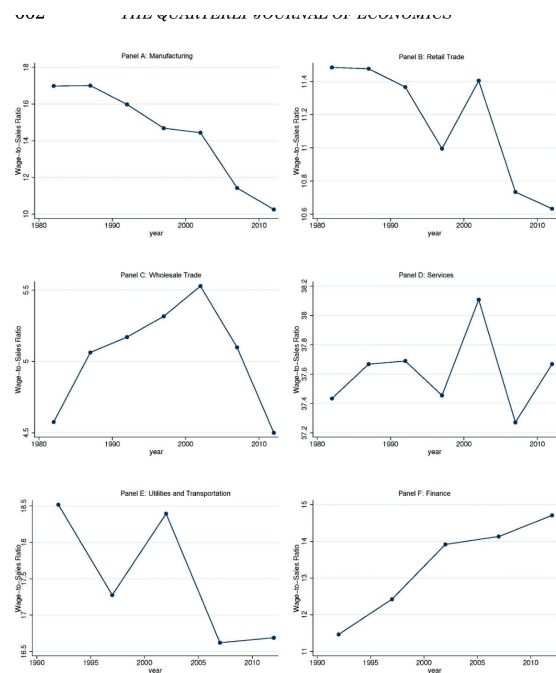


FIGURE III

Average Payroll-to-Sales Ratio

Each panel plots the overall payroll-to-sales ratio in one of the six major sectors covered by the U.S. Economic Census. These figures update Autor et al. (2017a) to include more recently released census data.

Figure III: payroll-to-sales ratio by sector

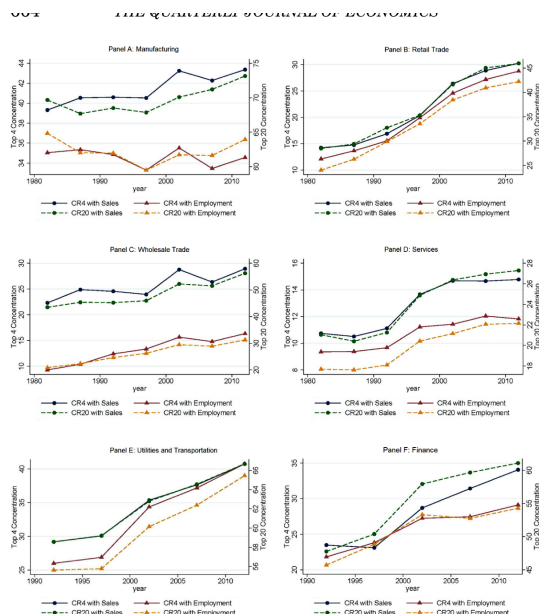


FIGURE IV

Average Concentration across Four-Digit Industries by Major Sector

This figure plots the average concentration ratio in six major sectors of the U.S. economy. Industry concentration is calculated for each time-consistent four-digit industry code, and then averaged across all industries within the six sectors. Each industry is weighted by its share of total sales within the sector. The solid blue line (circles), plotted on the left axis, shows the average fraction of total industry sales that is accounted for by the largest four firms in that industry, and the solid red line (triangles), also plotted on the left axis, shows the average fraction of industry employment used in the four largest firms in the industry. Similarly, the dashed green line (circles), plotted on the right axis, shows the average fraction of total industry sales that is accounted for by the largest 20 firms in that industry, and the dashed orange line (triangles), also plotted on the right axis, shows the average fraction of industry employment utilized in the 20 largest firms in the industry.

Figure IV: concentration by sector

7.3 Table I and Figure V: summary statistics and firm-size relationship

Read:

- Table I summarizes sector-level employment, firms, labor-share changes, and concentration levels and changes.
- Manufacturing, retail, wholesale, services, finance, and utilities/transportation all show large firm counts and meaningful concentration variation.
- Figure V shows that larger firms have lower labor shares within sectors.

Economic message: Superstar firms are large and have low labor shares. As their market shares rise, aggregate labor shares fall.

	Establish- ments (1)	Firms (2)	Payroll to Sales (3)	Δ Payroll to Sales (4)	Payroll to Value (5)	Δ Payroll to Value (6)	CR4 (7)	Δ CR4 (8)	CR20 (9)	Δ CR20 (10)
Manufacturing	183,400	141,300	13.58	-0.92	31.65	-2.18	42.97	1.00	72.33	0.87
88 industries, 2,328 obs.)	(12,560)	(11,490)	(8.15)	(2.10)	(12.27)	(5.35)	(21.66)	(7.08)	(22.04)	(4.50)
Retail trade	1,499,000	1,001,000	11.25	-0.10			22.10	2.34	38.01	2.71
5 industries, 348 obs.)	(57,400)	(20,040)	(5.77)	(0.89)			(19.71)	(4.58)	(25.95)	(3.94)
Wholesale trade	391,000	305,500	5.007	0.05			24.62	0.79	47.92	1.68
3 industries, 336 obs.)	(12,850)	(14,310)	(3.27)	(0.84)			(11.91)	(6.82)	(16.97)	(6.75)
Services	2,058,000	1,744,000	36.12	-0.40			13.59	0.75	24.56	1.01
5 industries, 570 obs.)	(300,800)	(212,600)	(10.93)	(2.23)			(13.39)	(4.50)	(18.76)	(4.78)
Finance	675,600	434,500	12.88	0.85			28.21	1.79	57.84	3.32
1 industries, 124 obs.)	(71,480)	(43,430)	(8.55)	(3.58)			(13.53)	(6.57)	(17.55)	(6.10)
Utilities and transportation	291,100	192,500	17.27	-0.39			32.66	1.14	61.44	1.17
5 industries, 144 obs.)	(18,560)	(6,545)	(8.23)	(2.39)			(20.90)	(7.39)	(22.38)	(5.80)

Notes: Summary statistics are based on the Economic Census of 1982-2012 for manufacturing, services, wholesale trade, and retail trade, and 1992-2012 for finance and utilities transportation. In manufacturing, we observe 388 consistently defined industries during six periods, and thus have $6 \times 388 = 2,328$ observations. Columns (1) and (2) indicate number of establishments and number of firms, and reflect totals for the entire sector, with the standard deviation across years in parentheses. Columns (3)-(10) indicate the level and five-year changes in payroll-to-sales, payroll-to-value added in manufacturing, and CR4 or CR20 sales concentration. These sector-level variables are based on weighted ranges of the underlying four-digit industries within a sector, where the weight is the industry's share of sales in the initial year when a sector is first covered by our data.

Table I: summary statistics

7.4 Table II and Table III: concentration and labor-share regressions

Read:

- Table II reports manufacturing regressions of labor-share changes on concentration changes.
- Rising CR4, CR20, and HHI are associated with falling labor share.
- Robustness controls include value added, trade exposure, industry trends, and initial conditions.
- Table III extends the relationship across sectors.
- The negative relationship appears in manufacturing, retail, wholesale, services, utilities/transportation, finance, and pooled specifications.

Economic message: Industries with faster concentration growth experience larger labor-share declines.

	5-year changes			10-year changes		
	CR4 (1)	CR20 (2)	HHI (3)	CR4 (4)	CR20 (5)	HHI (6)
baseline	-0.148***	-0.228***	-0.213**	-0.132***	-0.153***	-0.165**
	(0.036)	(0.043)	(0.085)	(0.040)	(0.055)	(0.093)
Compensation share	-0.177***	-0.266***	-0.256**	-0.139**	-0.151**	-0.185
	(0.045)	(0.056)	(0.110)	(0.053)	(0.071)	(0.125)
Product service intermediates	-0.339***	-0.514***	-0.502***	-0.261***	-0.353***	-0.303
	(0.064)	(0.074)	(0.173)	(0.066)	(0.085)	(0.275)
Tom value added in labor share	-0.219***	-0.337***	-0.320***	-0.210***	-0.251***	-0.289***
	(0.028)	(0.045)	(0.060)	(0.047)	(0.054)	(0.075)
concentration	-0.172***	-0.280***	-0.243**	-0.186**	-0.249***	-0.229*
	(0.043)	(0.047)	(0.100)	(0.059)	(0.088)	(0.128)
industry trends	-0.187***	-0.309***	-0.281**	-0.136***	-0.156***	-0.165**
	(0.043)	(0.061)	(0.102)	(0.040)	(0.055)	(0.093)
four-digit dummies	-0.163***	-0.285***	-0.233**	-0.148***	-0.168***	-0.181**
	(0.036)	(0.052)	(0.089)	(0.040)	(0.055)	(0.093)
1992-2012 subperiod	18.809***	20.467***	20.957***	-0.122***	-0.148***	-0.161*
	(3.027)	(3.213)	(3.187)	(0.040)	(0.053)	(0.092)
Control for initial capital	-0.148***	-0.231***	-0.214**	-0.148***	-0.168***	-0.181**
	(0.035)	(0.042)	(0.084)	(0.040)	(0.053)	(0.092)
Control for initial capital	-1.242***	-1.285***	-1.278***	-2.535***	-2.648***	-2.669***
	(0.308)	(0.324)	(0.292)	(0.595)	(0.598)	(0.565)
Employment-based	0.036	0.024	0.160**	0.018	0.029	0.082
	(0.036)	(0.033)	(0.073)	(0.035)	(0.040)	(0.083)
concentration measure						

Notes: $N = 2,328$ (88 industries $\times$ 6 five-year periods) in columns (1)-(3) (except $N = 1,502$ in rows 6 and 7) and $N = 1,164$ (38 industries $\times$ 3 10-year periods) in columns (4)-(6). It cell displays the coefficient from a separate OLS industry-level regression of the change in labor share on period fixed effects and the change in the concentration measure (at the top of each column). Industries are weighted by their total value added in the initial year, and standard errors in parentheses are clustered by four-digit industries. Models in rows 2 and 3 replace the baseline outcome variable (the change in payroll divided by the ratio of total compensation to value added, and payroll to value added net of intermediate services, respectively. Row 4 replaces the baseline regression (the change in value added concentration) with the change in concentration of value added. Row 5 uses concentration measures based on employment. Row 6 suggests the baseline model with a full set of four-digit industry dummies and thus controls for four-digit industry fixed effects in each industry. Rows 7 and 8, respectively, extend the baseline specification with controls for the change in the ratio of imports to value added, and the initial level of capital value added. Since the import measure is only available since 1992, the row (7) model is estimated only for five-year changes during the 1992-2012 period, while row 8 indicates in-sample from the baseline regression for this shorter period. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table II: manufacturing regressions

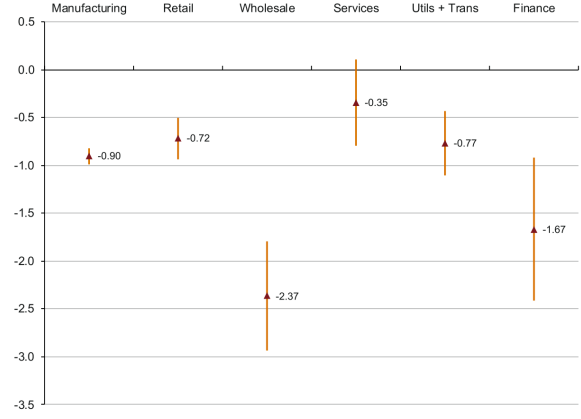


FIGURE V

The Relationship between Firm Size and Labor Share

The figure indicates OLS regression estimates that relate the level of a firm's labor share (payroll-to-sales ratio) to its share of overall sales in its four-digit industry. The six sector-specific regressions include all years available for that sector and control for year fixed effects. Industries are weighted by their sales in the initial year. Dots indicate coefficient estimates and lines indicate 95% confidence intervals based on standard errors clustered at the four-digit industry level.

Figure V: firm size and labor share

TABLE III
INDUSTRY REGRESSIONS OF THE CHANGE IN THE PAYROLL-TO-SALES RATIO ON THE CHANGE IN CONCENTRATION, DIFFERENT SECTORS

	Stacked 5-year changes			Stacked 10-year changes		
	CR4 (1)	CR20 (2)	HHI (3)	CR4 (4)	CR20 (5)	HHI (6)
1 Manufacturing	-0.062***	-0.077***	-0.112***	-0.035	-0.034	-0.088**
$n = 2,328; 1,164$	(0.013)	(0.025)	(0.026)	(0.021)	(0.033)	(0.037)
2 Retail	-0.034*	-0.084**	-0.041	-0.043**	-0.067**	-0.068***
$n = 348; 174$	(0.020)	(0.037)	(0.025)	(0.018)	(0.029)	(0.023)
3 Wholesale	-0.038***	-0.040**	-0.084**	-0.037**	-0.036*	-0.064
$n = 336; 168$	(0.014)	(0.017)	(0.041)	(0.018)	(0.019)	(0.048)
4 Services	-0.091	-0.128***	-0.350***	-0.093	-0.137***	-0.377**
$n = 570; 258$	(0.057)	(0.039)	(0.084)	(0.070)	(0.042)	(0.156)
5 Utilities/Transport	-0.110***	-0.111**	-0.320***	-0.064	-0.096**	-0.226**
$n = 144; 48$	(0.031)	(0.050)	(0.082)	(0.044)	(0.038)	(0.098)
6 Finance	-0.221**	-0.252***	-0.567**	-0.236**	-0.274***	-0.723**
$n = 124; 62$	(0.084)	(0.091)	(0.208)	(0.095)	(0.084)	(0.295)
7 Combined	-0.077***	-0.088***	-0.150***	-0.060***	-0.076***	-0.118***
$n = 3,850; 1,901$	(0.017)	(0.022)	(0.028)	(0.018)	(0.023)	(0.032)

Notes: Numbers of observations ($n = x; y$) are indicated below each sector for the first three columns (x) and the last three columns (y). Each cell displays the coefficient from a separate OLS industry-level regression of the change in labor share (payroll-to-sales ratio) on period fixed effects and the change in the concentration measure indicated at the top of each column. Industries are weighted by their sales in the initial year, and standard errors in parentheses are clustered by four-digit industries. In manufacturing, retail, services, and wholesale, we pool data from 1982-2012 and in finance and utilities and transportation, we pool data from 1992-2012. The combined regression in row 7 includes six sector fixed effects. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table III: sector-level regressions

7.5 Figure VI and Figure VII: regression coefficients and decomposition

Read:

- Figure VI displays sector-specific coefficients linking changes in labor share to changes in concentration.
- The estimated coefficients are negative across all sectors.
- Figure VII decomposes the manufacturing labor-share decline using the Melitz-Polanec method.
- Survivor reallocation and firm exit explain much of the decline.

Economic message: The labor-share decline is a reallocation phenomenon: activity shifts toward low-labor-share firms.

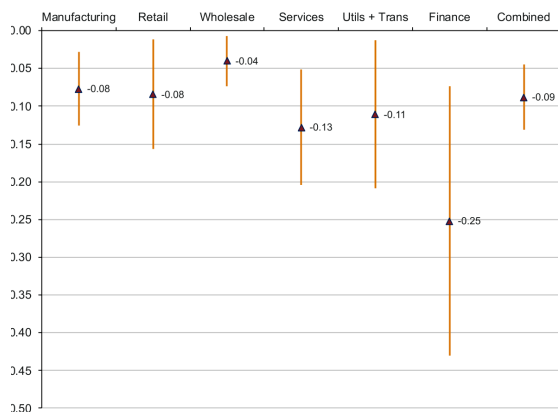


FIGURE VI

The Relationship between the Change in Labor Share and the Change in Concentration across Six Sectors

The figure indicates OLS regression estimates that relate Δ Labor Share (payroll over sales) to Δ CR20. The six sector-specific regressions include stacked five-year changes from 1982 to 2012 (1992 to 2012 in utilities/transportation and finance) and control for period fixed effects. Industries are weighted by their sales in the initial year. Dots indicate coefficient estimates, and lines indicate 95% confidence intervals based on standard errors clustered at the four-digit industry level. The

Figure VI: concentration coefficients by sector

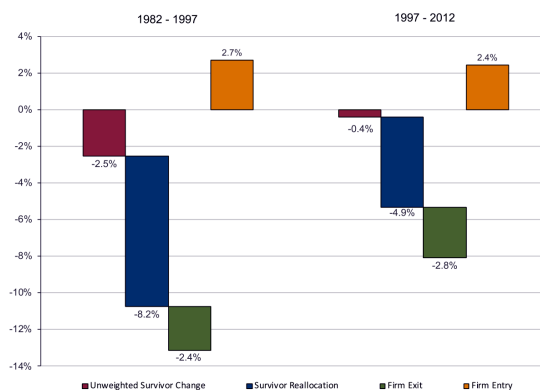


FIGURE VII

Melitz-Polanec Decomposition of the Change in Labor Share in Manufacturing

Each bar represents the cumulated sum of the Melitz-Polanec decomposition components calculated over adjacent five-year intervals for payroll over value added. The left side shows the sum of the decompositions from 1982–1987, 1987–1992, and 1992–1997 and the right side shows the sum of the decompositions from 1997–2002, 2002–2007, and 2007–2012. [Table IV](#) reports the underlying estimates for each five-year period.

Figure VII: Melitz-Polanec decomposition, manufacturing

7.6 Table IV and Figure VIII: decomposition across sectors

Read:

- Table IV shows the detailed manufacturing decomposition by five-year periods.
- Figure VIII shows cumulative Melitz-Polanec components across sectors.
- Reallocation among surviving firms is the largest negative contributor in most sectors.

Economic message: The reallocation mechanism generalizes outside manufacturing, consistent with the superstar-firm story.

TABLE IV
DECOMPOSITIONS OF THE CHANGE IN THE LABOR SHARE, MANUFACTURING

	Δ Un-weighted mean of survivors (1)	Survivor re-allocation (2)	Exit (3)	Entry (4)	Total (5)
Panel A: Payroll share of value added					
1982–1987	–3.03	–1.75	–0.59	0.86	–4.52
1987–1992	2.60	–5.26	–0.90	0.98	–2.58
1992–1997	–2.08	–1.24	–0.89	0.89	–3.32
1997–2002	0.00	–0.76	–1.00	0.69	–1.08
2002–2007	–3.06	–1.53	–1.12	1.23	–4.48
2007–2012	2.64	–2.61	–0.63	0.51	–0.09
1982–2012	–2.93	–13.15	–5.14	5.15	–16.07
Panel B: Compensation share of value added					
1982–1987	–0.78	–5.66	–0.47	0.98	–5.93
1987–1992	3.73	–5.69	–1.00	1.05	–1.91
1992–1997	–2.78	–1.90	–0.93	0.97	–4.64
1997–2002	–2.07	1.11	–1.09	0.79	–1.25
2002–2007	1.26	–6.21	–1.20	1.55	–4.60
2007–2012	0.40	–0.32	–0.77	0.53	–0.15
1982–2012	–0.24	–18.67	–5.46	5.89	–18.48

Notes. This table shows the results of a decomposition of the change in the labor share for the payroll share of value added in Panel A and for the compensation share of value added in Panel B using the dynamic Melitz and Polanec (2015) methodology as described in the text. We divide the change in the overall labor share (column (5)) into four components: column (1) indicates the change in the labor share due to a general decline across all surviving firms; column (2) captures reallocation among incumbent (surviving) firms due to the growing relative size of low labor share incumbent firms (and the interaction of the growth in their size and the growth in their labor share); columns (3) and (4), respectively, indicate the contribution of firm exit and

Table IV: manufacturing decomposition

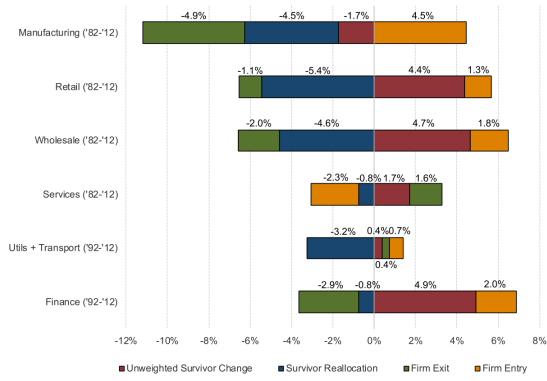


FIGURE VIII

Melitz-Polanec Decomposition of the Change in Labor Share in All Six Sectors

Each bar represents the cumulated sum of the Melitz-Polanec decomposition components calculated over adjacent five-year intervals for payroll over sales. Table V reports the underlying estimates for each five-year period.

last row shows, for example, that the compensation share fell by 18.5 percentage points between 1982 and 2012 and that essen-

Figure VIII: decomposition across sectors

7.7 Table V and Figure IX: reallocation components and concentration

Read:

- Table V reports detailed dynamic decompositions for all sectors and periods.
- Figure IX regresses decomposition components on changes in concentration.
- Concentration growth is most strongly related to the survivor-reallocation component.

Economic message: Concentrating industries are the industries where market-share reallocation contributes most to falling labor share.

TABLE V
DECOMPOSITIONS OF THE CHANGE IN THE PAYROLL-TO-SALES RATIO, ALL SECTORS

	Δ Un-weighted mean of survivors (1)	Survivor re-allocation (2)	Exit (3)	Entry (4)	Total (5)
Panel A: Manufacturing					
1982–1987	–0.73	0.68	–0.77	0.83	0.02
1987–1992	0.99	–1.92	–0.78	0.68	–1.02
1992–1997	–0.71	–0.51	–1.03	0.95	–1.30
1997–2002	0.50	–0.82	–0.68	0.75	–0.24
2002–2007	–2.24	–0.73	–0.94	0.90	–3.02
2007–2012	0.47	–1.24	–0.74	0.35	–1.17
1982–2012	–1.71	–4.54	–4.94	4.46	–6.73
Panel B: Retail					
1982–1987	–0.04	–0.19	0.19	–0.01	–0.01
1987–1992	–0.98	–0.20	0.15	–0.11	–1.14
1992–1997	–1.70	–0.24	0.27	–0.37	–2.04
1997–2002	–1.11	–0.08	0.18	0.41	–0.60
2002–2007	–0.94	–0.23	0.23	–0.67	–1.57
2007–2012	–0.67	–0.15	0.26	–0.10	–0.66
1982–2012	–5.44	–1.10	1.29	–0.85	–6.10
Panel C: Wholesale					
1982–1987	0.71	–0.40	–0.21	0.38	0.49
1987–1992	0.68	–0.52	–0.35	0.30	0.11
1992–1997	1.20	–1.03	–0.35	0.32	0.15
1997–2002	1.61	–1.39	–0.42	0.42	0.21
2002–2007	–0.29	–0.08	–0.30	0.25	–0.43
2007–2012	0.75	–1.16	–0.34	0.15	–0.60
1982–2012	4.66	–4.59	–1.97	1.82	–0.08
Panel D: Services					
1982–1987	0.56	0.83	–0.95	0.23	0.67
1987–1992	0.08	0.73	–0.64	0.02	0.19
1992–1997	–1.95	0.48	–0.12	–0.23	–1.82
1997–2002	0.02	0.22	0.05	0.65	0.94
2002–2007	0.20	–0.31	–0.49	–0.84	–1.44
2007–2012	0.33	–0.37	–0.15	0.40	0.21
1982–2012	–0.76	1.57	–2.30	0.23	–1.26

Table V, part 1: all-sector decomposition

TABLE V
CONTINUED

	Δ Un-weighted mean of survivors (1)	Survivor re-allocation (2)	Exit (3)	Entry (4)	Total (5)
Panel E: Utilities and Transportation					
1992–1997	1.14	–2.38	–0.34	0.34	–1.24
1997–2002	0.35	0.20	0.48	0.09	1.12
2002–2007	–0.98	–1.19	0.21	0.18	–1.78
2007–2012	–0.13	0.12	0.00	0.08	0.07
1992–2012	0.37	–3.25	0.36	0.69	–1.83
Panel F: Finance					
1992–1997	1.10	0.22	–0.66	0.30	0.96
1997–2002	1.65	–0.02	–0.70	0.57	1.50
2002–2007	1.51	–1.49	–0.45	0.64	0.21
2007–2012	0.66	0.54	–1.09	0.47	0.58
1992–2012	4.92	–0.75	–2.89	1.98	3.25

otes. This table shows the results of a decomposition of the change in the labor share using the dynamic Melitz and Polanec (2015) methodology as described in the text and notes table IV.

Table V, part 2: continuation

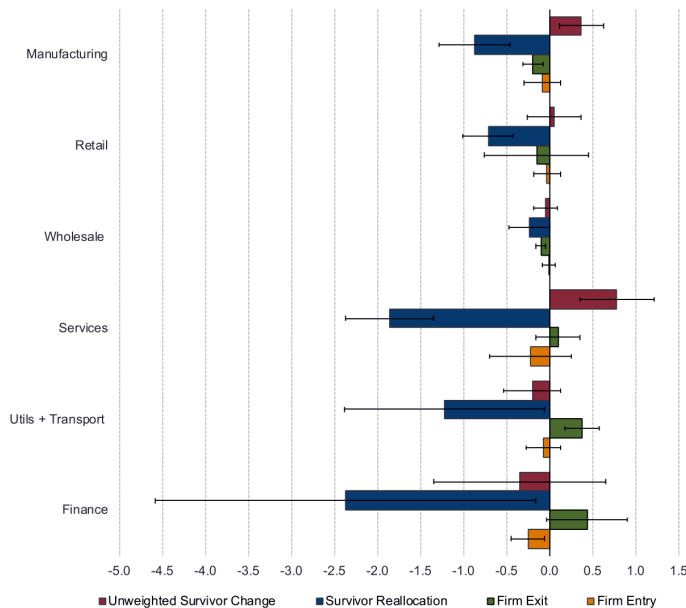


FIGURE IX
Regressions of the Components of the Change in Labor Share on the Change in Concentration

Figure IX: decomposition components versus concentration

7.8 Figure X and Table VI: markups and industry characteristics

Read:

- Figure X shows that aggregate markups rise over time in manufacturing across multiple measurement approaches.
- The weighted average markup rises more than the unweighted average or median markup.
- Table VI shows that concentrating industries have faster growth in patents per worker, value added per worker, TFP, and other productivity-related measures.

Economic message: Rising concentration is associated with productivity growth and reallocation toward high-markup firms, not merely with declining competition in a static sense.

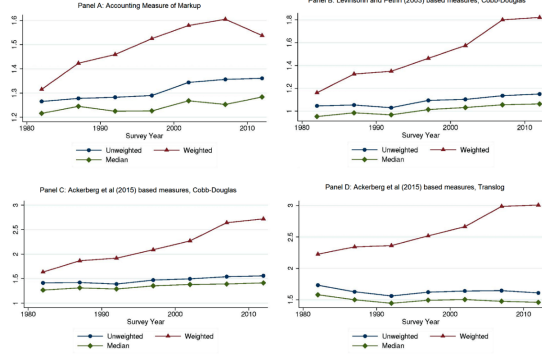


FIGURE X
Markup Changes

Each panel indicates estimates of the markup of price over marginal cost in manufacturing using the first-order condition described in the text (equation (7)). Panel A uses the Antràs et al. (2017) “accounting” method, and Panels B–D use production function methods following De Loecker and Warzynski (2012) where we estimate industry-specific production functions for two-digit SIC industries. In Panels B and C, the production function is assumed to be Cobb-Douglas, and

Figure X: markup changes

TABLE VI
CHARACTERISTICS OF CONCENTRATING INDUSTRIES

	CR4 (1)	CR20 (2)	HHI (3)
Panel A: Manufacturing only			
1 Patents per worker	0.090** (0.006)	0.057*** (0.022)	0.056** (0.022)
2 Value added per worker	0.126*** (0.028)	0.074*** (0.020)	0.067*** (0.025)
3 Capital per worker	0.092** (0.041)	0.026 (0.022)	0.081*** (0.029)
4 Five-factor TFP	0.055*** (0.019)	0.024* (0.013)	0.028* (0.017)
5 Payroll per worker	0.013 (0.018)	0.005 (0.011)	0.016 (0.010)
6 Material costs per worker	0.120*** (0.028)	0.074*** (0.018)	0.068*** (0.023)
Panel B: All sectors			
7 Manufacturing sales per worker	0.125*** (0.027)	0.067*** (0.018)	0.069*** (0.016)
8 Retail sales per worker	0.049 (0.048)	0.098 (0.067)	0.027 (0.023)
9 Wholesale sales per worker	0.16*** (0.058)	0.207*** (0.042)	0.031** (0.013)
10 Services sales per worker	0.082 (0.055)	0.125*** (0.036)	0.041** (0.019)
11 Utilities/transportation sales per worker	0.415*** (0.096)	0.304*** (0.092)	0.117*** (0.023)
12 Finance sales per worker	0.270* (0.143)	0.216* (0.111)	0.144*** (0.052)
13 Combined sales per worker	0.155*** (0.031)	0.147*** (0.026)	0.053*** (0.011)

Notes. $N = 2,328$ (388 industries $\times$ 6 five-year periods) in Panel A, and various sector-specific numbers of observations as reported in Table IV in Panel B. Each cell displays the coefficient from a separate OLS industry-level regression of the change in sales concentration on period fixed effects and the change in

Table VI: characteristics of concentrating industries

8 Short summary

- Labor’s share has fallen in the United States and many other countries.
- The paper proposes the rise of superstar firms as a unifying mechanism.
- Superstar firms are large, productive, high-markup, and low-labor-share firms.
- As sales shift toward superstar firms, aggregate labor share falls.
- U.S. Economic Census data show rising concentration across major sectors.
- Industries with bigger concentration increases experience larger labor-share declines.
- Decomposition evidence shows that reallocation among surviving firms is a major driver of the fall in labor share.
- Markup evidence shows that aggregate markups rise more than the typical firm’s markup, consistent with reallocation toward high-markup firms.
- International data show similar patterns between concentration and labor-share decline.
- The paper shifts the debate from factor substitution alone to firm heterogeneity and product-market structure.

9 Conclusion

9.1 Core conclusion

Autor, Dorn, Katz, Patterson, and Van Reenen argue that the fall in the labor share is closely connected to the rise of superstar firms. The most productive and dominant firms in an industry have relatively low labor shares. When more sales flow to these firms, aggregate labor share declines even if average firm behavior changes little.

9.2 Economic intuition

The superstar mechanism is a reallocation mechanism. Technological change, globalization, and scale economies can make consumers more likely to buy from the most productive firms. Those firms grow, concentration rises, aggregate markups rise, and labor's share falls because production shifts toward firms that use labor less intensively relative to value added.

9.3 Takeaway

The paper's main lesson is that aggregate income shares cannot be understood without firm heterogeneity. The identity of firms producing output matters. The decline in labor's share is therefore not just a macro factor-price phenomenon, but also a micro reallocation and product-market-structure phenomenon.